

phishing Awareness Analysis

Introduction

Cybersecurity plays an important role in protecting digital systems and user information from cyber threats. One of the most common cyber attacks is phishing. Phishing is a social engineering attack where attackers attempt to trick users into revealing sensitive information such as usernames, passwords, banking details, and personal data through fake emails, messages, or websites.

The purpose of this project is to analyze phishing emails and identify suspicious elements that indicate malicious intent. This project helps improve cybersecurity awareness and teaches users how to recognize and avoid phishing attacks.

Objective of the Project

The main objectives of this project are:

- To understand phishing attacks
- To identify suspicious emails and fake links
- To analyze phishing techniques used by attackers
- To improve cybersecurity awareness
- To learn safe browsing and communication practices

Tools and Technologies Used

- Internet Browser
- Sample phishing emails/messages
- GitHub for project upload
- MS Word / Google Docs for documentation
- Python (optional for phishing detection)

Sample Phishing Message

Dear Customer,

Your bank account has been temporarily suspended due to unusual activity. Please verify your account immediately using the link below to avoid permanent account closure.

<http://secure-bank-login-verification.xyz>

Thank you,

Customer Support Team

Analysis of the Message

The above email contains several suspicious elements that indicate a phishing attempt. Attackers use fear and urgency to manipulate users into clicking malicious links and entering confidential information.

Red Flags Identified

1. Suspicious URL

The link provided in the email is not an official banking domain. Attackers often create fake websites that resemble legitimate services.

2. Urgent Language

The message creates panic using phrases like:

- “Immediately verify”
- “Permanent account closure”

This pressures users into acting without thinking.

3. Request for Sensitive Information

Legitimate banks never ask users to verify passwords or personal details through email links.

4. Unknown Sender

The sender address may look unofficial or slightly modified to imitate a trusted company.

Example:

support@bank-security-alert.xyz

5. Grammar and Formatting Issues

Many phishing emails contain spelling mistakes or unusual sentence structures.

Why the Message is Unsafe

The message is unsafe because it attempts to redirect users to a fake website where sensitive information can be stolen. Such attacks may lead to:

- Identity theft
- Financial loss
- Unauthorized account access
- Malware infection

Users should avoid clicking unknown links and verify messages before responding.

Prevention Methods

To stay protected from phishing attacks:

- Never click unknown links
- Verify sender email addresses
- Use two-factor authentication
- Avoid downloading suspicious attachments
- Keep software updated
- Report phishing emails immediately
- Use antivirus protection

Skills Learned

Through this project, the following skills were developed:

- Threat Analysis
- Cybersecurity Awareness
- Security Thinking
- Risk Identification
- Email Analysis
- Phishing Detection Techniques

Python Code for Detection

```
keywords = ["urgent", "verify", "password", "bank", "click"]
```

```
message = input("Enter message: ").lower()
```

```
for word in keywords:
```

```
    if word in message:
```

```
        print("Possible phishing attempt detected!")
```

```
        break
```

```
else:
```

```
    print("Message appears safe.")
```

Output

Possible phishing attempt detected!

Conclusion

This project provided practical knowledge about phishing attacks and cybersecurity awareness. By analyzing suspicious messages and identifying red flags, users can better protect themselves from online threats. The project also improved analytical thinking and understanding of cyber attack methods used by hackers.

References

- Cybersecurity Awareness Articles
- Online Phishing Examples
- GitHub Documentation
- Security Awareness Training Materials